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71.

Solution :

Given :

$\Delta H_C (CH_4)$ = – 890 kJ mol^{-1}

$\Delta H_C (C_2H_4)$ = –1423 kJ mol^{-1}

Let the mixture contain x lit of CH_4 and (3.67 – x) lit of ethylene.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

x lit x lit

$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

(3.67–x)lit 2 (3.67 – x) lit

Volume of Carbondioxide formed = x + 2 (3.67–x) = 6.11 lit

x + 7.34 – 2x = 6.11

7.34 – x = 6.11

x = 1.23 lit

Given mixture contains 1.23 lit of methane and 2.44 lit of ethylene, hence

$$\Delta H_C = \left[\frac{\Delta H_C (CH_4)}{22.4 \text{ lit}} \times (x) \text{ lit}\right] + \left[\frac{\Delta H_C (C_2H_4)}{22.4 \text{ lit}} \times (3.67 - x) \text{ lit}\right]$$

$$\Delta H_C = \left[\frac{-890 \text{ kJ mol}^{-1}}{22.4 \text{ lit}} \times 1.23 \text{ lit}\right] + \left[\frac{-1423}{22.4 \text{ lit}} \times (3.67 - 1.23) \text{ lit}\right]$$

$\Delta H_C = [-48.87 \text{ kJ mol}^{-1}]$

$+ [-155 \text{kJ mol}^{-1}]$

$\Delta H_C = -203.87 \text{ kJ mol}^{-1}$